

MODULE 6

Conclusions and future scope

6.1 Conclusion

The Smart Event Management and Ticket Booking System aims to create a streamlined, organized, and user-friendly platform to manage events and online ticket bookings efficiently. It's a web application that includes event management, booking, payment, user and authentication.

The information for the events are managed by admin in the Event Management Module; users can access and browse available events. Bookings Management Module: It offers a handy way for users to choose events and reserve tickets. The Payment Management Module stores payment information linked to bookings, and allows for the simulation of a payment process for educational purposes.

User Management and Authentication Module facilitates secure login and registration of user and administrator. Data is encrypted using bcrypt and protected routes/resources are secured with JWT authentication. Role based authorization helps to make sure that admin functionalities are only available to authorized users.

The system information is stored and managed in the cloud, with MongoDB Atlas selected as the cloud database. The design of the frontend was done with React.js and Vite, and the backend REST APIs were built with Node.js and Express.js. The full application has been subjected to unit testing, integration testing and system testing to guarantee appropriate working and interaction amongst the various parts.

The whole application was successfully deployed by Vercel in the front and Railway in the back, supported by the database services of MongoDB Atlas. This shows that the system can be deployed as a full-fledged application in the cloud and isn't just a system for local development.

In conclusion, the project meets the main goal of building a single place for event management and ticketing, with all important aspects including secure authentication, data organization, and user-friendliness being carried out effectively.

6.2 Advantages

The Smart Event Management and Ticket Booking System offers a number of benefits:

A one-stop solution for event management and ticketing.
Helps users easily navigate and book events.
Eliminates manual booking and records.
Provides secure user and administrator authentication.
Utilises passwords that are encrypted to enhance security of accounts.
Provides access to admin functions according to roles.
Remains efficient with the booking and payment system.
Relies on the cloud-based data storage service MongoDB Atlas.
Offers a friendly and responsive user interface.
Once it is deployed, it gives access via internet.
Saves administrative challenges by using digital administration.
Offers future functionality in a scalable manner.

6.3 Limitations

Even though the system will contain only the essentials, there are some limitations because of the project's scope and academic background.

Simulated payment method and not based on a real payment gateway.
Real-time payment verification is not in place.
Currently, no integration of email and SMS notifications.
QR Ticket Verification is not available.
Advanced event analytics and reporting is limited.
No mobile app with a focus has been created.
Currently there are no advanced recommender services provided.
The system deployed requires a network connection.
These large scale load and stress tests have not been carried out.

These are restrictions that can be overcome in future versions of system.

6.4 Future Scope

There are a number of features that can be added to the Smart Event Management and Ticket Booking System in the future.

Real Payment Gateway Integration

Real payment gateway like Razorpay or Stripe can be used in place of simulated one. This would enable safe online settlement and instant confirmation on the transactions.

Generate QR Code tickets.

Custom QR Codes can be created for successful reservations. These codes can be scanned at event entrances to check tickets and prevent duplicate and unauthorized entries.

Get mobile and instant notifications via email and SMS.

Email and SMS' can be incorporated to send sign up confirmations, booking confirmations, payment receipts, reminders, and updates on the event.

Use advanced event search and filtering.

The advanced search feature can be added based on events category, location, date, price, popularity, and availability helping the users find the right events.

Seat Selection

For events that allow users to more than catch up on seats prior to booking, a visual seat selection system can be implemented.

Event Analytics

Advanced analytics such as ticket sales, revenue, booking trends, event popularity and user activity can be given to administrators.

Mobile Application

A dedicated Android and iOS app can be created, which will make it easier for the users to discover, book, pay for, and access the events using their digital tickets.

AI-Based Recommendations

AI can be used to suggest events based on the user's interests, past bookings, browsing history, and interests.

Real-Time Notifications

You can use the push notification to offer users up-to-date information about changes, cancellation, booking state and upcoming events.

Multi-Language Support

It can be expanded to allow multiple languages to be used so it is accessible to a wider number of people.

Advanced Administration

Future releases include complex administrative tools like detailed reports, automatic revenue calculation, event performance analysis, monitoring user movement and configure system functions in one place.

6.5 References / Bibliography

The following resources are referred while building Smart Event Management and Ticket Booking System:

1. React.js Documentation
2. Vite Documentation
3. Node.js Documentation
4. Express.js Documentation
5. MongoDB Documentation
6. Mongoose Documentation
7. Bootstrap Documentation
8. JSON Web Token Documentation
9. bcrypt Documentation
10. Git Documentation
11. GitHub Documentation
12. Vercel Documentation
13. Railway Documentation

6.6 Final Project Outcome

Smart Event Management and Ticket Booking System fully illustrates a modern full stack web app implementation. The project encompasses frontend developing, backend API building, database administration, authentication, event management, booking

management, payment method administration, testing and cloud deployment adding together in a built-in application.

With the system fully in place, future enhancements can be made for a full-fledged event management system, including real-time payment processing, digital ticket verification, notifications, analytics, mobile apps, and AI-driven recommendations.